

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #1

Activity Name: GPIO LED Interfacing

Name of Student:	AGNEAY B NAIR
Registration No.:	CH.SC.U4CSE24102
Activity Date:	31-07-27
Submission Date:	31-07-27

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	10	

Faculty In-charge

Write an embedded C program for ARM architecture to interface an LED. Debug and verify the program using IDE on a development computer system.

```
#include<LPC213x.h>

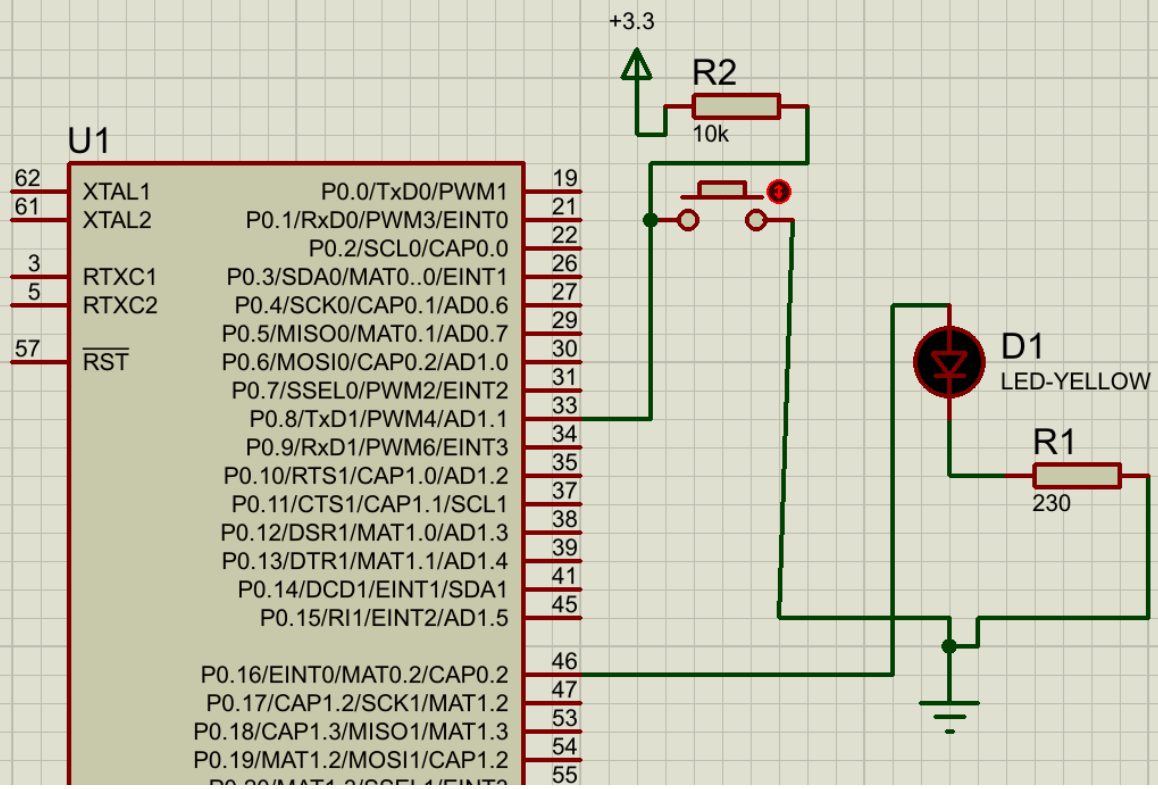
void delay(int n){
    int i;
    for(i = 0;i<n*1000;i++);
}

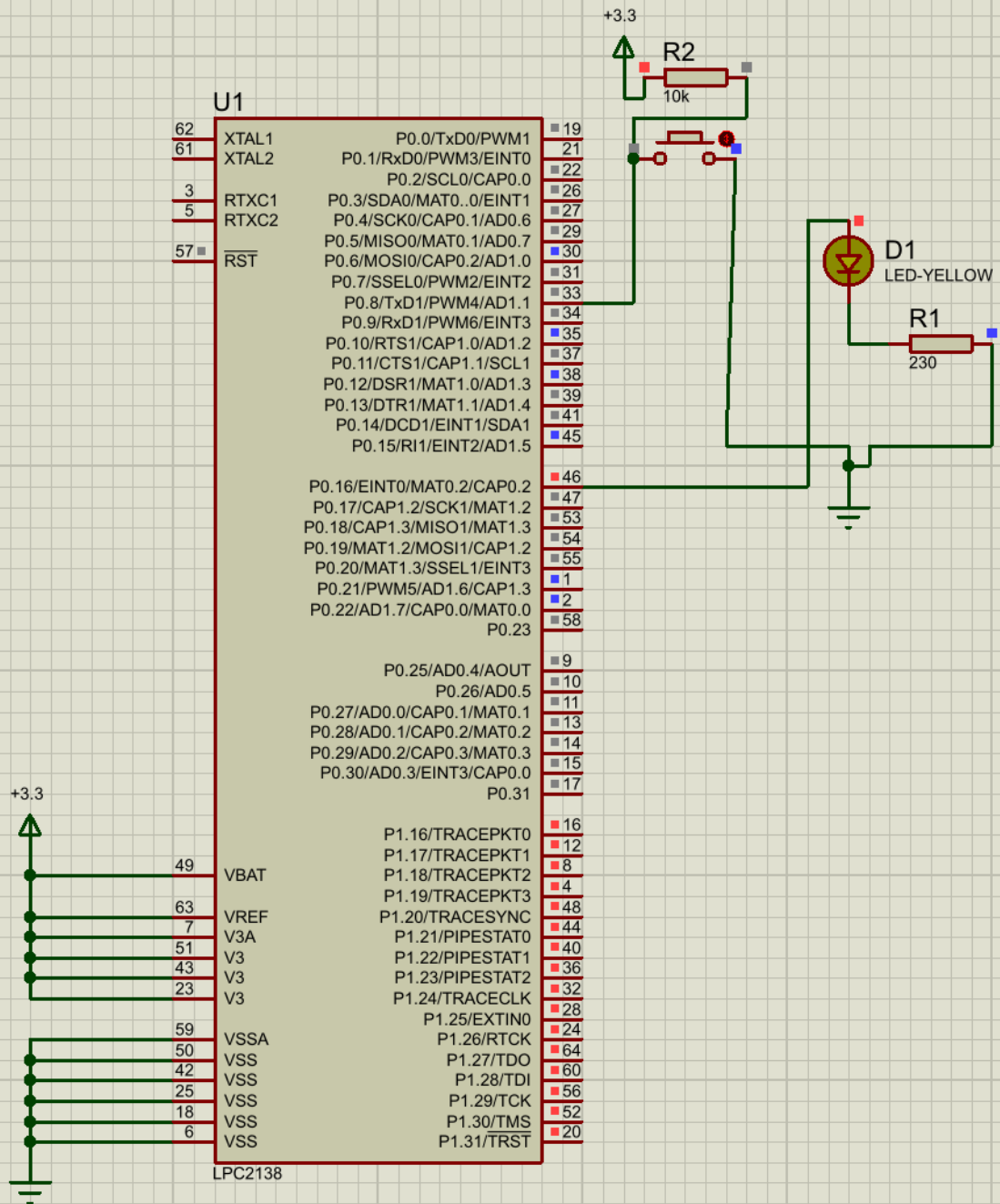
int main(){
    PINSEL0 = 0x00000000;
    IODIR0 |= (1<<16);
    IODIR0 &= ~(1<<8);

    while(1){
        if(IOPIN0 & (1<<8)){
            // turn LED on
            IOSET0 = (1<<16);
            delay(10);
        }else{
            IOCLR0 = (1<<16);
            delay(10);
        }
    }
}
```

```
1  #include<LPC213x.h>
2
3  void delay(int n){
4      int i;
5      for(i = 0;i<n*1000;i++);
6  }
7
8  int main(){
9      PINSEL0 = 0x00000000;
10     IODIR0 |= (1<<16);
11     IODIR0 &= ~(1<<8);
12
13     while(1){
14         if(IOPIN0 & (1<<8)){
15             // turn LED on
16             IOSET0 = (1<<16);
17             delay(10);
18         }else{
19             IOCLR0 = (1<<16);
20             delay(10);
21         }
22     }
23 }
```

OUTPUT:





C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 1\Code\First Program.uvproj - µVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
R0	0x00000000
R1	0x00000000
R2	0x00000000
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12 (SP)	0x00000000
R14 (LR)	0x00000000
R15 (PC)	0x00000000
CPSR	0x00000000
SPSR	0x00000000
User/System	
Fast Interrupt	
Interrupt	
Supervisor	
Abort	

Disassembly

9: PINSEL0 = 0x00000000;

```
0x000001BC E3A00000 MOV R0,#0x00000000
0x000001C0 E59F106C LDR R1,[PC,#0x006C]
0x000001C4 F5810000 STR R0,[R1]
```

FirstC

```
5 for(i = 0;i<n*1000;i++);
6 }
7
8 int main(){
9     PINSEL0 = 0x00000000;
10    IODIR0 |= (1<<16);
11    IODIR0 &= ~(1<<8);
12
13    while(1){
14        if(IOPIN0 & (1<<8)){
15            // turn LED on
16            IOSET0 = (1<<16);
17            delay(10);
18        }else{
```

Command

Load "C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 1\Code\

*** Restricted Version with 32768 Byte Code Size Limit

*** Currently used: 588 Bytes (1%)

Call Stack + Locals

Name	Locati...	Type
n	0x0000...	int f0

ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess

Call Stack + Locals Memory 1

Real-Time Agent: Target Reset Simulation t1: 0.00007217 sec L:9 C:1 CAP: NUM SCRL OVR: R/W